Computer Networks & TCP/IP Protocol Architecture (3150710)

Uploaded By: Harsh Shah (5th Sem Information Technology) | Official Course Curriculum | Peer Verified & 4.9

Unit 1: OSI 7-Layer Model vs TCP/IP Suite

• Physical Layer: Bit transmission, Manchester encoding, modulation.

• Data Link Layer: Framing, Error detection (CRC-32, Checksum), Flow control (Sliding Window, Go-Back-N, Selective Repeat).

• Network Layer: Logical IP addressing, packet forwarding, routing (OSPF, BGP, RIP).

• Transport Layer: Process-to-process delivery (TCP reliable connection-oriented vs UDP unreliable connectionless).

• Application Layer: HTTP/2, DNS, SMTP, FTP, WebSocket.

Unit 2: IPv4 / IPv6 Subnetting & Addressing Numericals

• Classless Inter-Domain Routing (CIDR): /24 = 256 IPs, /26 = 64 IPs (62 usable hosts).

• Variable Length Subnet Masking (VLSM) formula: 2^(32 - prefix) - 2 usable host addresses.

• Subnetting Solved Example: Network 192.168.10.0/24 split into 4 subnets: Subnet 0: 192.168.10.0/26, Subnet 1: 192.168.10.64/26, Subnet 2: 192.168.10.128/26, Subnet 3: 192.168.10.192/26.

Unit 3: TCP Connection & Congestion Control

• TCP 3-Way Handshake: SYN -> SYN-ACK -> ACK. Termination via FIN -> ACK -> FIN -> ACK.

• TCP Congestion Control Algorithms: Slow Start, Congestion Avoidance (Additive Increase Multiplicative Decrease - AIMD), Fast Retransmit, Fast Recovery.